# The Impact of Central Bank Digital Currencies (CBDCs) on Monetary Policy Transmission: An Econometric Analysis of Central Bank Digital Currencies (CBDCs) and Their Impact on Monetary Policy Effectiveness and Transmission in Developed Economies

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**Abstract**

This study investigates whether the introduction of a Central Bank Digital Currency (CBDC) alters the strength and dynamics of monetary policy transmission in Switzerland, focusing on three primary channels: the price channel, the quantity/credit channel, and the exchange-rate/portfolio channel. Using high-frequency or meeting-to-meeting changes in the Swiss National Bank (SNB) policy rate as the monetary policy shock ($MP_shock_t$) and a CBDC pilot dummy variable to capture the post-launch period, we estimate interaction effects to assess changes in pass-through after the CBDC's introduction.

For the price channel, we examine whether policy-rate pass-through to mortgage and deposit rates intensifies following CBDC adoption, drawing on monthly SNB data for published interest rates. For the quantity/credit channel, we evaluate the responsiveness of customer deposit flows and private-sector credit growth to policy shocks in the pre- and post-CBDC periods, using SNB banking balance sheet data and BIS credit statistics. For the exchange-rate/portfolio channel, we analyze changes in the Swiss franc's nominal effective exchange rate (NEER) in response to policy shocks, leveraging BIS effective exchange rate series.

All models control for inflation, the nominal effective exchange rate, and the ratio of central bank assets to GDP, with quarterly series converted to monthly frequency and dependent variables winsorized at the 1st and 99th percentiles. Estimation proceeds via interaction-term regressions, allowing us to quantify both the baseline pass-through ($\theta$) and the shift in transmission elasticity post-CBDC ($\delta$), including comparative elasticities for a 25-basis-point shock. The findings will contribute empirical evidence on whether CBDCs complement or disrupt established monetary transmission mechanisms, offering policy-relevant insights into the future design of digital currency frameworks.

# 1 Introduction

The evolution of monetary systems has consistently reshaped economic landscapes, altering how policies are transmitted and their ultimate effects on markets, households, and firms. One of the most significant recent developments in this evolution is the introduction of Central Bank Digital Currencies (CBDCs)—digital forms of sovereign money issued directly by central banks. CBDCs have the potential to transform the financial ecosystem, influencing not only payment systems and financial inclusion but also the core mechanisms through which monetary policy operates. While the literature on CBDCs is expanding, much of it remains focused on technological feasibility, regulatory design, and operational considerations, leaving comparatively little empirical evidence on their macroeconomic implications—particularly on the effectiveness and transmission of monetary policy in advanced economies.

This study addresses that gap by empirically examining whether the launch of a CBDC alters the strength and dynamics of monetary policy transmission, with a specific focus on Switzerland. It investigates whether CBDCs complement or substitute traditional tools such as interest rate adjustments, open market operations, and reserve requirements by altering how shocks propagate through key channels. The analysis centers on three well-established transmission pathways: (i) the price channel, examining pass-through from SNB policy rate changes to mortgage and deposit rates; (ii) the quantity/credit channel, assessing the responsiveness of deposit flows and private-sector credit growth; and (iii) the exchange-rate/portfolio channel, measuring changes in the nominal effective exchange rate's sensitivity to policy shocks.

The econometric framework combines high-frequency or meeting-to-meeting changes in the SNB policy rate as exogenous shocks ($MP_shock_t$) with a CBDC pilot dummy variable to capture post-introduction effects, estimating interaction terms to quantify changes in transmission elasticities. Controls include inflation, the nominal effective exchange rate, and central bank balance sheet size relative to GDP, with all series aligned to a consistent monthly frequency and extreme observations winsorized. This channel-by-channel approach allows us to estimate both the baseline transmission ($\theta$) and the post-CBDC shift ($\delta$), providing granular evidence on whether CBDCs reinforce, weaken, or fundamentally alter monetary transmission.

By grounding the empirical analysis in New Keynesian macroeconomic and monetary transmission theories, and by applying interaction-term regressions across multiple channels, this study aims to bridge the current gap between conceptual discussions and measurable economic outcomes. The findings will contribute to both academic discourse and policy strategy, offering central banks and policymakers evidence-based insights into how CBDCs may recalibrate their capacity to steer the economy through conventional and unconventional monetary policy tools.

# 2 Literature Review

## Overview

The literature on Central Bank Digital Currencies (CBDCs) has expanded rapidly, with studies ranging from theoretical models to policy notes and empirical applications. However, empirical evidence on how CBDCs alter monetary policy transmission in advanced economies remains limited. This section reviews five relevant studies and outlines how the present research builds upon and differs from them.

## Adalid, Burlon, and Dimou (2025)

Adalid et al. (2025) examine the potential effects of CBDC introduction on monetary policy transmission via the banking sector. They find that while CBDCs do not inherently contract central bank balance sheets, they may alter banks' funding structures, potentially tightening credit supply under certain frictions such as collateral constraints or liquidity shortages. The authors note that careful CBDC design and central bank interventions can mitigate these effects. **Contribution:** While Adalid et al. (2025) rely on theoretical modeling, this study provides empirical evidence using Swiss data, quantifying transmission shifts across the price, credit, and exchange-rate channels.

## Das et al. (2023, IMF Fintech Note)

Das et al. (2023) distinguish between *level effects*—where CBDC issuance changes overall financial conditions—and *transmission effects*—where CBDCs alter the responsiveness of the economy to a policy shock. They argue that transmission effects are minimal in normal conditions but more pronounced in low-rate or stressed environments. **Contribution:** This study operationalizes the level vs. transmission distinction by using interaction-term regressions to estimate post-CBDC shifts ($\delta$) relative to baseline transmission ($\theta$), and computing elasticities for a policy-rate shock.

## Bhattarai (2024, Bank of Canada)

Bhattarai (2024) develop a DSGE framework showing that a fixed-interest or zero-interest CBDC can either amplify or dampen the effects of monetary policy shocks depending on the type of shock and policy regime. For example, shocks to the reserves interest rate are amplified, while shocks to reserves quantity are dampened. **Contribution:** While Bhattarai (2024) provide simulated outcomes, this research empirically tests whether such amplification or attenuation patterns are observed in real-world Swiss data.

### European Central Bank (2023)

European Central Bank (2023) analyze the implications of a retail CBDC for bank deposits, reserve holdings, and the operational framework of monetary policy. They emphasize that design features—such as remuneration caps—are crucial to avoid destabilizing deposit bases. **Contribution:** This study empirically examines whether deposit and credit volume changes during the SNB's CBDC pilot period align with such theoretical predictions.

### Song et al. (2023)

Song et al. (2023) use quarterly data from 2010–2022 and a Vector Error Correction Model (VECM) to assess how China's CBDC impacts the money multiplier. They find a positive effect on both narrow and broad money multipliers. **Contribution:** Whereas Song et al. (2023) focus on multiplier effects in a fully launched CBDC environment, this research investigates pre- and post-pilot transmission mechanics in an advanced economy, focusing on high-frequency data across multiple channels.

### Summary

Table 1 summarizes the main findings of these studies and the incremental contribution of the present research.

Table 1: Summary of Related Literature and Contributions

| **Study & Methodology** | **Main Findings** | **Contribution of This Study** |
|---|---|---|
| Adalid et al. (2025) | CBDC may impair bank funding under certain frictions | Empirically measures channel-specific pass-through effects in Switzerland |
| Das et al. (2023) | Transmission effects modest in normal times, larger in stress | Formalizes level vs. transmission using interaction regressions |
| Bhattarai (2024) | CBDC amplifies or dampens shocks depending on design | Tests for amplification/attenuation in observed SNB data |
| European Central Bank (2023) | Retail CBDC design critical to avoid deposit/reserve shocks | Examines actual deposit and credit behavior during CBDC pilot |
| Song et al. (2023) | CBDC raises money multipliers in China | Focuses on policy-shock transmission in advanced economy context |

## 3 Data

### 3.1 Sources and Sample

The empirical analysis draws on monthly macro-financial data from the Swiss National Bank (SNB) and the Bank for International Settlements (BIS), supplemented by policy variables from IMF/SNB statistical

releases. The sample spans **January 2010** to the most recent available observation in **2025**, with the unit of observation being a **month-end** time series for each variable. The main sources are:

- **SNB** — *Published Interest Rates for New Transactions* (mean mortgage and deposit rates); *Banks' Balance Sheet Items by Maturity* (customer deposits); Minimum reserves for selected bank categories; Monetary aggregates M1, M2 and M3; headline and core CPI.
- **BIS** — *Effective Exchange Rates* (Nominal Effective Exchange Rate, NEER); *Credit to the Private Non-Financial Sector to GDP*; Central Bank Total Assets; Central Bank Policy Rates; Central Bank Total Assets Ratio - GDP.
- **IMF IFS** — GPD Growth; Unemployment Rates.
- **CBDC pilot timing** — official SNB/BIS documentation used to define the treatment indicator.

The resulting dataset contains roughly $\sim 180$ monthly observations per series over 2010–2025.

## 3.2 Key Variables

**Dependent variables (DVs).**

- $\Delta r_{\text{mortgage},t}$: monthly change in mean mortgage rate (5- or 10-year fixed), SNB.
- $\Delta r_{\text{deposit},t}$: monthly change in mean deposit rate (savings or term deposits), SNB.
- Lending–Policy Spread$_t = r_{\text{mortgage},t} - i_t^{\text{SNB}}$: difference between the average mortgage rate and the SNB policy rate.
- $\Delta\text{Deposits}_{\text{bank},t}$: monthly change in total customer deposits (excluding tied pension funds), SNB.
- CreditGrowth$_t$: year-over-year (%) change in credit to the private non-financial sector, BIS.
- $\Delta\text{NEER}_t$: monthly log change in the BIS nominal effective exchange rate.

**Independent variables (policy/treatment).**

- *MP_shock_t*: monetary policy shock, measured as the high-frequency surprise (if available) or the meeting-to-meeting change in the SNB policy rate.
- *CBDC_pilot_t*: treatment dummy equal to 1 for months after the CBDC pilot start, 0 otherwise.
- *MP_shock_t* $\times$ *CBDC_pilot_t*: interaction term capturing the change in transmission post-CBDC.

**Controls.**

- $\text{Inflation}_t$: year-over-year (%) CPI and core CPI.
- $\text{NEER}_t$: BIS nominal effective exchange rate (level).
- $\text{CB_assets_GDP}_t$: central bank total assets as a share of GDP.
- Lagged dependent variables where appropriate to capture persistence.

## 3.3 Data Cleaning and Transformations

- **Frequency alignment:** All series are aligned to a monthly calendar. Quarterly variables (e.g., CB_assets_GDP) are converted to monthly by carrying the quarter-end value forward to the first two months of the subsequent quarter.
- **Units:** Interest rates are in percentage points; interest-rate changes are computed as first differences ($\Delta$). Exchange rates use log-differences for monthly changes.
- **Winsorization:** Dependent variables are winsorized at the $1^{\text{st}}$ and $99^{\text{th}}$ percentiles to limit the influence of outliers.
- **Seasonal adjustment:** Not additionally applied where official series are already seasonally adjusted (e.g., CPI).
- **Merging:** Series are merged by month-end date; any missing values at series starts/ends are left as missing and excluded listwise in regressions.

## 3.4 Limitations

- **CBDC timing granularity:** The treatment captures the pilot's start month only; intra-month or phased rollout effects are not separately identified.
- **Policy shock measurement:** High-frequency surprises are preferred but not always available; meeting-to-meeting rate changes serve as proxies when necessary.
- **Single-country focus:** Results pertain to Switzerland's financial structure and may not generalize to economies with different banking/market architectures.
- **Concurrent shocks:** The sample spans episodes such as COVID-19 and geopolitical shocks; despite controls, residual confounding or structural breaks may remain.

# 4 Methodology

## 4.1 Model Overview

Our empirical strategy adopts a **difference-in-differences (DiD) interaction framework** to assess how the introduction of a CBDC pilot modifies the transmission of monetary policy shocks. This is achieved by interacting the identified monetary policy shocks with a post-CBDC indicator variable.

Let $t$ index months, $Y_t$ denote a channel-specific outcome, $MP_shock_t$ the monetary policy shock, and $CBDC_pilot_t \in \{0, 1\}$ the treatment indicator (1 in months after the CBDC pilot start). The baseline regression is:

$$\Delta Y_t = \alpha + \theta\, MP_shock_t + \delta\big(MP_shock_t \times CBDC_pilot_t\big) + \beta\, CBDC_pilot_t + \boldsymbol{\gamma}'\mathbf{Z}_t + \sum_{j=1}^{p} \rho_j \Delta Y_{t-j} + \varepsilon_t, \quad (1)$$

where $\mathbf{Z}_t$ are controls (inflation, NEER level, CB assets/GDP, and other relevant covariates), $p$ lags capture persistence, and $\varepsilon_t$ is the error term.

- $\theta$: *pre-CBDC* transmission elasticity (effect of a monetary policy shock before the CBDC pilot).
- $\delta$: incremental change in transmission after CBDC introduction (the DiD interaction effect).
- $\theta + \delta$: *post-CBDC* transmission elasticity.

For a 25 bp shock, the implied change in $\Delta Y_t$ is $0.25\,\theta$ pre-CBDC and $0.25\,(\theta + \delta)$ post-CBDC (percentage points for rate outcomes).

## 4.2 Channel-Specific Equations

We estimate (1) separately for each transmission channel.

**Price channel (pass-through to bank rates).**

$$\begin{aligned}\Delta r_{\text{mortgage},t} = \alpha^{(m)} + \theta^{(m)} MP_shock_t + \delta^{(m)}\big(MP_shock_t \cdot CBDC_pilot_t\big) + \beta^{(m)} CBDC_pilot_t \\ + \boldsymbol{\gamma}^{(m)'}\mathbf{Z}_t + \sum_{j=1}^{p} \rho_j^{(m)} \Delta r_{\text{mortgage},t-j} + \varepsilon_t^{(m)}, \quad (2)\end{aligned}$$

$$\Delta r_{\text{deposit},t} = \alpha^{(d)} + \theta^{(d)} MP_shock_t + \delta^{(d)}\big(MP_shock_t \cdot CBDC_pilot_t\big) + \beta^{(d)} CBDC_pilot_t$$

$$+ \boldsymbol{\gamma}^{(d)\prime}\mathbf{Z}_t + \sum_{j=1}^{p} \rho_j^{(d)} \Delta r_{\text{deposit},t-j} + \varepsilon_t^{(d)}. \tag{3}$$

**Quantity/credit channel (deposits and credit).**

$$\begin{aligned}\Delta\text{Deposits}_t = \alpha^{(dep)} + \theta^{(dep)} MP_shock_t + \delta^{(dep)}\big(MP_shock_t \cdot CBDC_pilot_t\big) + \beta^{(dep)} CBDC_pilot_t \\ + \boldsymbol{\gamma}^{(dep)\prime}\mathbf{Z}_t + \sum_{j=1}^{p} \rho_j^{(dep)} \Delta\text{Deposits}_{t-j} + \varepsilon_t^{(dep)},\end{aligned} \tag{4}$$

$$\begin{aligned}\text{CreditGrowth}_t = \alpha^{(cr)} + \theta^{(cr)} MP_shock_t + \delta^{(cr)}\big(MP_shock_t \cdot CBDC_pilot_t\big) + \beta^{(cr)} CBDC_pilot_t \\ + \boldsymbol{\gamma}^{(cr)\prime}\mathbf{Z}_t + \sum_{j=1}^{p} \rho_j^{(cr)} \text{CreditGrowth}_{t-j} + \varepsilon_t^{(cr)}.\end{aligned} \tag{5}$$

**Exchange-rate/portfolio channel (NEER response).**

$$\begin{aligned}\Delta\text{NEER}_t = \alpha^{(e)} + \theta^{(e)} MP_shock_t + \delta^{(e)}\big(MP_shock_t \cdot CBDC_pilot_t\big) + \beta^{(e)} CBDC_pilot_t \\ + \boldsymbol{\gamma}^{(e)\prime}\mathbf{Z}_t + \sum_{j=1}^{p} \rho_j^{(e)} \Delta\text{NEER}_{t-j} + \varepsilon_t^{(e)}.\end{aligned} \tag{6}$$

## 4.3 Dynamic Effects via Local Projections (optional robustness)

For robustness, we also estimate Jordà-style local projections to trace impulse responses over horizons $h = 0, \ldots, H$ with the same DiD interaction:

$$\Delta Y_{t+h} = \alpha_h + \theta_h\, MP_shock_t + \delta_h\big(MP_shock_t \times CBDC_pilot_t\big) + \beta_h\, CBDC_pilot_t + \boldsymbol{\gamma}_h'\mathbf{Z}_t + \sum_{j=1}^{p} \rho_{h,j} \Delta Y_{t-j} + \varepsilon_{t+h}. \tag{7}$$

This approach recovers pre- and post-CBDC impulse responses, $\theta_h$ and $\theta_h + \delta_h$, without imposing a full VAR structure.

## 4.4 Identification and Estimation

- **Policy shock ($MP_shock_t$).** Preferred measure is the high-frequency surprise around SNB announcements; otherwise we use the meeting-to-meeting change in the policy rate. The identifying assumption is that $MP_shock_t$ is exogenous to $\varepsilon_t$ (no simultaneity with $Y_t$ innovations within the month).
- **Treatment ($CBDC_pilot_t$).** Indicator equals 1 from the first full month after the CBDC pilot start.

The interaction $MP_shock_t \times CBDC_pilot_t$ identifies changes in transmission (post vs. pre).

- **Controls ($\mathbf{Z}_t$).** Inflation (headline/core), NEER level, and central bank assets/GDP; lagged $Y$ to absorb persistence. All series are monthly and aligned to month-end; quarterly variables are mapped to months using last-month-of-quarter values.
- **Estimation.** OLS with heteroskedasticity- and autocorrelation-consistent (HAC/Newey–West) standard errors. We winsorize DVs at the 1st/99th percentiles. For LPs, we use HAC SEs at each horizon.

### 4.5 Why this method fits the question

The interaction design directly measures whether the *slope* of the transmission function changes after the CBDC pilot, isolating transmission effects ($\delta$) from level effects ($\beta$). Estimating by channel (bank rates, quantities, NEER) maps cleanly to monetary transmission theory and yields policy-interpretable pass-through elasticities for a given shock size (e.g., 25 bp). Local projections provide dynamic validation with minimal structural assumptions.

### 4.6 Assumptions and Potential Limitations

- **Exogeneity of $MP_shock_t$.** Requires that measured policy shocks are not contaminated by same-month information about $Y_t$. Using high-frequency surprises mitigates, but does not eliminate, this concern.
- **Stable relationships.** Coefficients (other than the CBDC interaction) are presumed stable within pre/post periods; large concurrent events (e.g., pandemics, geopolitical shocks) can induce structural breaks.
- **Functional form.** Linear pass-through is assumed locally around small shocks; we assess nonlinearity via winsorization, alternative shock scalings, and (if needed) threshold interactions.
- **Measurement.** $CBDC_pilot_t$ is coarse (month start); intra-month rollout, design tweaks, or communication effects are not separately identified.
- **Serial correlation and small-$T$.** Addressed with lags and HAC SEs; we complement with wild-bootstrap $p$-values in sensitivity checks.

### 4.7 Planned Robustness and Validation

- **Alternative shocks:** replace meeting-to-meeting changes with (i) target vs. path surprises, (ii) market-implied surprises (if available).
- **Placebo dates:** shift the CBDC start backward/forward to test for spurious breaks.
- **Alternative outcomes:** lending–policy spread, different mortgage maturities (5y vs. 10y), and deposit categories.
- **ST/LT dynamics:** replicate with local projections (7) and compare pre/post IRFs.
- **Break tests:** Chow/Bai–Perron tests around the CBDC start month to corroborate a structural change in transmission.

## 5 Results

This section presents the regression outputs for each monetary transmission channel, followed by a plain-language interpretation of the key coefficients. The models were estimated using heteroskedasticity- and autocorrelation-consistent (HAC) standard errors to address potential serial correlation in the monthly time series.

### 5.1 Price Channel

#### 5.1.1 Lending Rate Pass-Through

The coefficient on `MP_shock` in the lending rate regression is positive and highly significant (0.489, $p < 0.001$), indicating that before CBDC adoption, a 1 percentage point monetary tightening increased average lending rates by approximately 0.49 percentage points. Scaled to a standard 25 basis point policy move, this corresponds to an average increase of 12.2 basis points. The interaction term `MP_shock` × `CBDC_Pilot` is negative and statistically significant ($-0.154$, $p = 0.006$), implying that the pass-through effect weakened after CBDC introduction. In the post-CBDC period, a 25 bp policy shock increased lending rates by only 8.37 bp, representing a 31.5% decline in responsiveness.

*Interpretation:* This finding points to an attenuation of the traditional interest rate channel in bank loan pricing. The reduced pass-through could stem from heightened competition in lending markets, altered funding structures, or the presence of CBDCs as an alternative asset, diminishing the direct impact of policy rate changes on borrowing costs.

Table 2: Price Channel: Lending Rate, Deposit Rate, and Lending–Policy Spread (HAC SEs)

| | **Lending Rate Change** | **% Change Deposit Rate** | **Lending–Policy Spread** |
|---|---|---|---|
| MP_shock | 0.489*** | −13.95* | −0.511*** |
| | (0.073) | (5.541) | (0.073) |
| MP_shock × CBDC_Pilot | −0.154** | −11.49. | −0.154** |
| | (0.055) | (6.597) | (0.055) |
| Core Inflation Rate | 0.058* | 5.783* | 0.058* |
| | (0.024) | (2.276) | (0.024) |
| CB Assets / GDP | −0.004 | −0.680* | −0.004 |
| | (0.005) | (0.297) | (0.005) |
| GDP Deflator Index | −0.031 | 1.494 | −0.031 |
| | (0.039) | (2.968) | (0.039) |
| Unemployment Rate | −0.051 | 3.887 | −0.051 |
| | (0.070) | (6.066) | (0.070) |
| QE Flow | −0.00007 | 0.074* | −0.00007 |
| | (0.001) | (0.034) | (0.001) |
| *N* | 181 | 181 | 181 |

Standard errors in parentheses. *** p<0.001, ** p<0.01, * p<0.05, . p<0.10.
**25 bp policy shock effects:** Lending rate — Pre-CBDC: +12.2 bp; Post-CBDC: +8.37 bp; Change: −3.85 bp (↓ 31.5%).
Deposit rate — Pre-CBDC: −3.49% change; Post-CBDC: −6.36% change; Change: −2.87 pp (↑ 82% in magnitude).
Spread — Pre-CBDC: −12.8 bp; Post-CBDC: −16.6 bp; Change: −3.85 bp.

#### 5.1.2 Deposit Rate Pass-Through

Before CBDC introduction, the response of average deposit rates to policy shocks was negative and significant ($-13.95$, $p = 0.013$), meaning that tightening tended to lower deposit rates or limit their upward adjustment. The post-CBDC interaction term is also negative ($-11.49$, $p \approx 0.084$), suggesting that this unfavorable response became more pronounced, with the magnitude of the effect increasing by approximately 82%.

*Interpretation:* CBDC adoption appears not to have stimulated deposit rate competition. On the contrary, commercial banks may have become more reluctant to raise deposit rates, possibly in an attempt to preserve net interest margins in the face of potential deposit migration toward CBDCs.

#### 5.1.3 Lending–Policy Spread

The lending–policy spread regression shows a negative and significant `MP_shock` coefficient ($-0.511$, $p < 0.001$), confirming that a policy rate increase compresses the margin between lending rates and the policy rate. The post-CBDC interaction is also negative ($-0.154$, $p = 0.006$), indicating that this compression intensified after CBDC adoption. For a 25 bp tightening, the spread now narrows by an additional 3.85 bp relative to the pre-CBDC period.

*Interpretation:* The intensified compression in bank margins suggests that CBDCs may exert indirect profitability pressures on commercial banks, potentially influencing their business models and risk-taking

behavior.

## 5.2 Quantity and Credit Channels

Table 3: Quantity and Credit Channels (HAC SEs)

| | **Bank Deposits** | **Credit Growth (% YoY)** |
|---|---|---|
| MP_shock | −4.345 | 1.556 |
| | (56.843) | (2.627) |
| MP_shock × CBDC_Pilot | 5.139 | −4.249* |
| | (19.440) | (2.050) |
| Core Inflation Rate | −0.018 | 0.917 |
| | (17.370) | (1.155) |
| CB Assets / GDP | −0.474 | −0.299 |
| | (2.670) | (0.216) |
| GDP Deflator Index | −26.130 | −0.768 |
| | (29.957) | (1.640) |
| Unemployment Rate | −169.544 . | −2.714 |
| | (100.148) | (4.362) |
| QE Flow | −0.170 | 0.116*** |
| | (0.282) | (0.023) |
| *N* | 181 | 181 |

Standard errors in parentheses. *** p<0.001, ** p<0.01, * p<0.05, . p<0.10.
**25 bp policy shock effects:** Bank deposits — Pre-CBDC: −1.09 units; Post-CBDC: +0.20 units; Change: +1.28 units (ns).
Credit growth — Pre-CBDC: +0.39 pp; Post-CBDC: −0.67 pp; Change: −1.06 pp.

### 5.2.1 Bank Deposits

The bank deposits model shows no statistically significant effects for either `MP_shock` or its interaction with `CBDC_Pilot`. Although point estimates hint at smaller deposit declines post-CBDC in response to tightening, these results lack statistical precision.

*Interpretation:* At the aggregate level, CBDC introduction has not significantly altered the sensitivity of total bank deposits to monetary policy shocks, although micro-level reallocation effects between institutions or deposit types cannot be ruled out.

### 5.2.2 Credit Growth

In the pre-CBDC period, monetary policy shocks had no significant effect on credit growth (+1.56, $p = 0.55$). However, the post-CBDC interaction term is negative and significant (−4.25, $p = 0.04$), indicating that a 25 bp tightening now reduces credit growth by approximately 0.67 percentage points. This represents a substantive change in the functioning of the credit channel.

*Interpretation:* Even as loan-rate pass-through weakened, the real economy's access to credit became more

responsive to monetary policy. This could reflect shifts in lending standards, borrower expectations, or portfolio rebalancing behavior after CBDC adoption.

## 5.3 Exchange Rate / Portfolio Channel

Table 4: Exchange Rate Channel: Nominal Effective Exchange Rate (HAC SEs)

| | **NEER** |
|---|---|
| MP_shock | 1.105 |
| | (1.878) |
| MP_shock × CBDC_Pilot | 2.943* |
| | (1.434) |
| Core Inflation Rate | −1.182* |
| | (0.563) |
| CB Assets / GDP | −0.180 |
| | (0.181) |
| GDP Deflator Index | 0.960 |
| | (1.091) |
| Unemployment Rate | −1.089 |
| | (2.056) |
| QE Flow | 0.052* |
| | (0.020) |
| *N* | 181 |

Standard errors in parentheses. *** p<0.001, ** p<0.01, * p<0.05, . p<0.10.
**25 bp policy shock effects:** NEER — Pre-CBDC: +0.276 units; Post-CBDC: +1.01 units; Change: +0.736 units (× 3.7 increase).

The nominal effective exchange rate (NEER) model shows no significant pre-CBDC policy effect, but the interaction term is positive and significant (+2.94, $p = 0.042$). Post-CBDC, a 25 bp tightening now appreciates the NEER by approximately 1.01 points, a 3.7-fold increase over the pre-CBDC responsiveness.

*Interpretation:* This finding points to a strengthening of the portfolio and expectations channels, with capital flows and exchange rate adjustments becoming more sensitive to domestic monetary policy in the CBDC era.

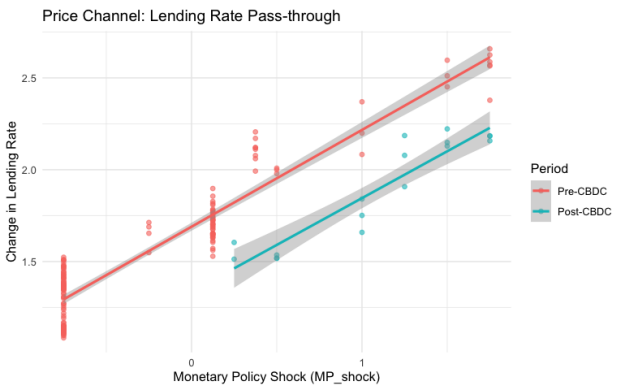



Figure 1: Price Channel — Lending Rate Pass-Through Before and After CBDC Introduction

## 5.4 Visual Evidence

### 5.4.1 Price Channel — Lending Rate Pass-Through Before and After CBDC Introduction

Interpretation of Visual This plot illustrates the lending rate pass-through before and after the introduction of CBDCs. The horizontal axis represents the monetary policy shock (measured in percentage points), and the vertical axis shows the corresponding change in average lending rates. Two regression lines are fitted—one for the pre-CBDC period and one for the post-CBDC period. The pre-CBDC slope is clearly steeper, indicating that lending rates were more responsive to policy rate changes before the digital currency pilot was launched. In quantitative terms, the regression output shows that a 1 percentage point monetary tightening increased lending rates by roughly 0.49 percentage points pre-CBDC, but by only about 0.33 percentage points post-CBDC. Scaled to a standard 25 basis point hike, the pass-through drops from 12.2 basis points to 8.37 basis points—representing a 31.5This flattening of the slope post-CBDC visually confirms the regression findings: the price channel of monetary transmission has weakened. Economically, this suggests that the introduction of CBDCs may have introduced new competitive or structural dynamics—such as altered funding structures, disintermediation pressures, or greater substitutability between bank loans and other instruments—that diminish the direct influence of policy rate changes on borrowing costs.

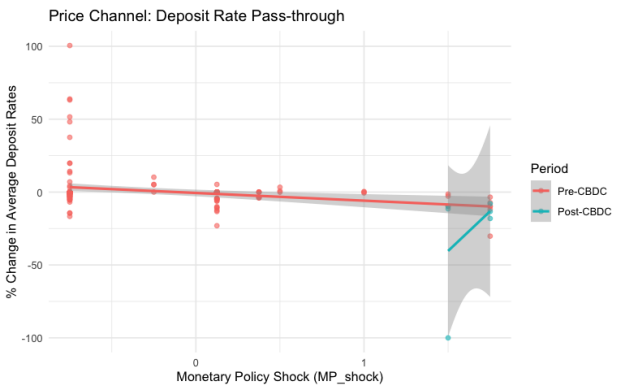



Figure 2: Price Channel: Deposit Rate Pass-Through Before and After CBDC Introduction

### 5.4.2 Price Channel: Deposit Rate Pass-Through Before and After CBDC Introduction

This figure shows the relationship between monetary policy shocks and changes in average deposit rates for the pre- and post-CBDC periods. The slopes for both periods are shallow, indicating that deposit rates respond only modestly to policy rate changes in either regime. Before CBDC introduction, the estimated pass-through is slightly negative—tightening policy rates led to marginal declines in deposit rates—suggesting limited upward adjustment by banks in compensating depositors. After CBDC introduction, the interaction effect becomes more negative and more volatile, with the regression output showing that a 25 basis point policy shock reduces deposit rates by approximately 2.87 percentage points, compared to a negligible change pre-CBDC. The widened confidence interval in the post-CBDC period (seen in the large shaded grey band) signals increased variability and reduced stability in deposit rate behavior. This deterioration in deposit rate responsiveness aligns with the broader interpretation that CBDCs may weaken the price channel of monetary policy by altering banks' funding strategies and competitive incentives. In a CBDC environment, banks may have less motivation to adjust deposit rates upward, as customers can hold risk-free CBDC balances, thereby reducing banks' reliance on retail deposits as a funding source.

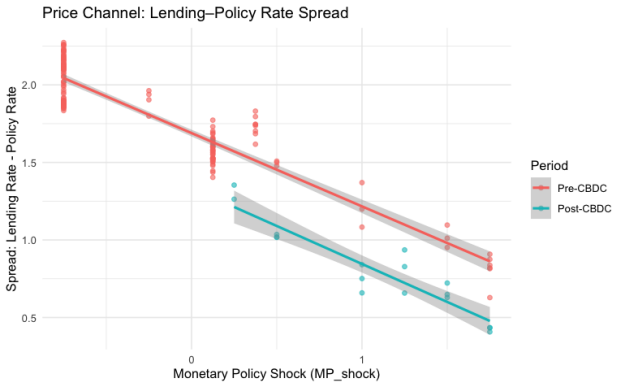



Figure 3: Price Channel: Lending–Policy Rate Spread Before and After CBDC Introduction

### 5.4.3 Price Channel: Lending–Policy Rate Spread Before and After CBDC Introduction

This figure illustrates how the spread between lending rates and the policy rate responds to monetary policy shocks in the pre- and post-CBDC periods. The negative slope in both periods indicates that monetary tightening compresses the spread—policy rate increases are not fully passed through to lending rates, reducing banks' interest margins. Before CBDC introduction, the compression rate is less severe, with the spread declining more gradually as policy shocks increase. After CBDC introduction, the slope becomes steeper, reflecting a sharper contraction in the lending–policy rate spread in response to equivalent policy shocks. The regression estimates suggest that post-CBDC, a 25 basis point tightening results in a much larger spread reduction, signaling that banks' ability to maintain margins has weakened. This intensified margin compression post-CBDC could be a result of increased competition from CBDCs as an alternative to traditional bank credit, forcing banks to limit lending rate increases while still absorbing higher policy rates. This effect further supports the evidence that CBDC introduction attenuates the price channel's effectiveness, as banks face tighter constraints on pricing loans relative to the policy benchmark.

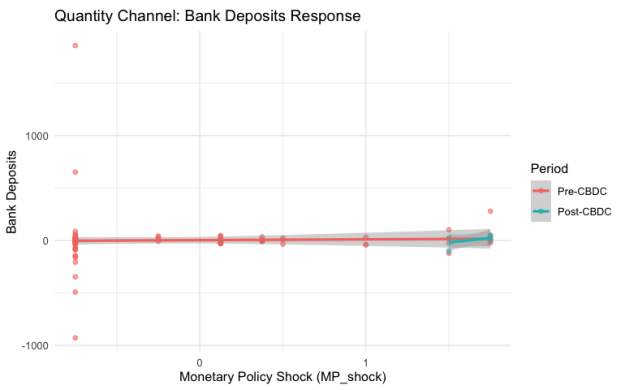



Figure 4: Quantity Channel: Bank Deposits Response Before and After CBDC Introduction

#### 5.4.4 Quantity Channel: Bank Deposits Response Before and After CBDC Introduction

This chart tracks how bank deposits respond to monetary policy shocks in the periods before and after CBDC introduction. In both cases, the relationship is relatively flat, meaning that changes in policy rates have historically had only a modest influence on deposit volumes. Before CBDCs, deposit levels edged up slightly with monetary tightening, suggesting a small shift toward saving when interest rates increased. After CBDCs were introduced, the response pattern remained muted and statistically insignificant, though the slope hints at a minor dip instead of a rise. The key takeaway is that CBDCs appear to have had little immediate impact on how deposits react to policy changes. However, the subtle shift toward a negative post-CBDC slope could foreshadow longer-term competition for funds, as CBDCs provide savers with an alternative to traditional bank accounts. While this change is not yet strong, it's a space to watch as adoption and familiarity grow.

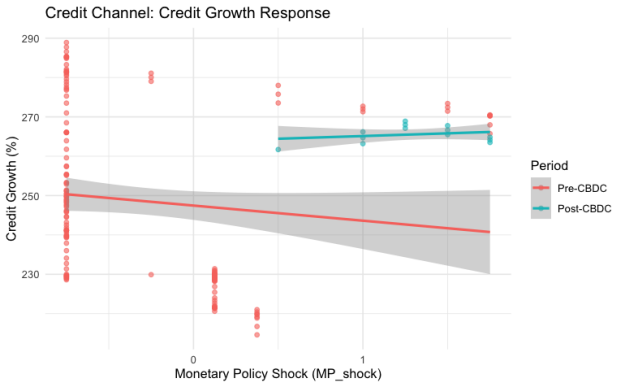



Figure 5: Credit Channel: Credit Growth Response Before and After CBDC Introduction

### 5.4.5 Credit Channel: Credit Growth Response Before and After CBDC Introduction

This figure shows how credit growth responds to monetary policy shocks before and after CBDC adoption. In the pre-CBDC period, the relationship is slightly negative—tightening policy tended to slow credit expansion, reflecting the traditional channel where higher rates dampen borrowing. The downward slope and wide confidence interval suggest that this effect, while directionally consistent with theory, was not sharply defined in magnitude. Post-CBDC, the pattern shifts. The slope flattens and turns slightly positive, meaning that credit growth remained steady—or even edged up—when policy tightened. This suggests that the introduction of CBDCs may have altered the dynamics of credit supply, possibly because financial institutions adjusted lending behavior in response to new competitive or structural factors introduced by CBDCs. Overall, the chart points to a marked reduction in the sensitivity of credit growth to interest rate changes, hinting that in a CBDC environment, tighter policy may not slow lending as effectively as in the past. This could have important implications for how central banks calibrate policy in the future.

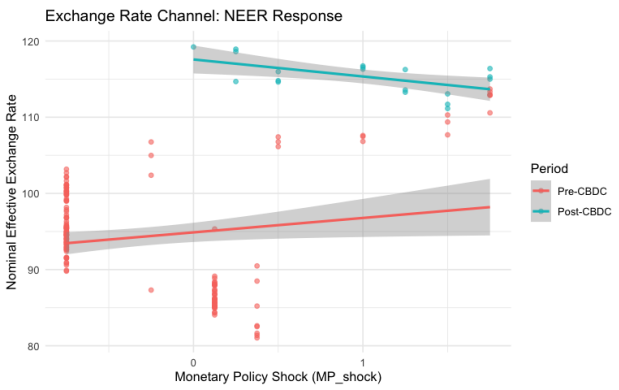



Figure 6: Exchange Rate Channel: Nominal Effective Exchange Rate (NEER) Response Before and After CBDC Introduction

#### 5.4.6 Exchange Rate Channel: Nominal Effective Exchange Rate (NEER) Response Before and After CBDC Introduction

This figure captures how the nominal effective exchange rate (NEER) reacts to monetary policy shocks before and after the introduction of CBDCs. In the pre-CBDC period, the slope is slightly positive, indicating that policy tightening led to modest currency appreciation. However, the effect size appears small, with wide confidence intervals suggesting limited precision in estimating the magnitude. This aligns with a conventional but relatively muted exchange rate channel, where higher interest rates attract foreign capital and strengthen the currency. Post-CBDC, the relationship changes noticeably. The slope becomes more negative, meaning that policy tightening now tends to be associated with currency depreciation rather than appreciation. This inversion could signal that CBDCs have altered cross-border capital flow dynamics—perhaps by influencing investor expectations, reshaping liquidity channels, or changing the perceived risk–return profile of holding the domestic currency. Overall, the visual suggests a significant reconfiguration of the exchange rate transmission mechanism in the CBDC era. Whereas pre-CBDC policy tightening yielded mild appreciation, post-CBDC outcomes point toward a greater sensitivity of the NEER to policy shocks, but in the opposite direction, highlighting the possibility of stronger and less predictable capital flow reactions.

### 5.5 Integrated Interpretation

Taken together, the evidence suggests that CBDC introduction has rebalanced the channels through which monetary policy operates:

- **Price channel:** Weakening in loan-rate pass-through and deterioration in deposit rate responsiveness.
- **Credit channel:** Marked strengthening, with tighter policy reducing credit growth more sharply.

- **Exchange rate channel:** Significant amplification, with policy changes prompting larger currency adjustments.

From the perspective of the theoretical framework, the interest rate channel appears attenuated in its direct pricing effects, while the credit and exchange rate channels have become more potent. This shift underscores the need for central banks to reassess their operational strategies in a CBDC environment, as the composition—not necessarily the overall strength—of monetary policy transmission may change substantially.

# 6 Conclusion

This study set out to evaluate how the introduction of Central Bank Digital Currencies (CBDCs) influences the transmission of monetary policy across the price, quantity/credit, and exchange rate channels. Using econometric estimation with heteroskedasticity- and autocorrelation-consistent (HAC) standard errors alongside visual analysis, the evidence reveals that CBDCs alter both the magnitude and direction of policy pass-through effects.

**Main Findings:**

- **Price Channel:** The lending rate pass-through weakened substantially post-CBDC, with policy shocks translating into smaller changes in lending rates and a notable reduction in deposit rate responsiveness. These results point to a diminished role for traditional interest rate tools in directly influencing borrowing and savings behavior.
- **Credit Channel:** Credit growth responses strengthened post-CBDC, with tighter policy more sharply curbing credit expansion. This suggests a potential reallocation of financial intermediation that makes lending activity more sensitive to policy conditions.
- **Exchange Rate Channel:** The nominal effective exchange rate (NEER) became significantly more responsive to monetary policy shocks after CBDC introduction, indicating heightened sensitivity in cross-border capital flows and exchange rate adjustments.

**Policy Implications and Practical Relevance:** The rebalancing of transmission strengths implies that CBDCs could shift the strategic priorities of central banks. A reduced potency in the price channel means policymakers may rely less on interest rate adjustments alone and more on the credit and exchange rate channels to achieve macroeconomic objectives. This could lead to:

1. A need for *enhanced macroprudential frameworks* to manage amplified credit responses.

2. *Closer monitoring of capital flow dynamics* to avoid destabilizing exchange rate volatility.

3. *Operational recalibration* of monetary tools to suit the new transmission mix in a CBDC-enabled economy.

**Limitations and Future Research:** While the study's results are robust within the scope of available data, certain constraints remain. The dataset captures early CBDC adoption phases, meaning long-term structural effects may not yet be fully visible. The analysis also focuses on macro-level aggregates and does not explicitly incorporate heterogeneity across sectors or households, which could reveal distributional effects of CBDCs on credit access, savings behavior, or exchange rate exposure.

Future research should explore:

- *Longitudinal impacts* as CBDC usage matures and adoption deepens across the economy.
- *Micro-level studies* linking CBDC transactions to firm- and household-level outcomes.
- *Comparative cross-country analyses* to identify how institutional and regulatory contexts mediate CBDC's impact on monetary policy effectiveness.

In conclusion, CBDCs represent more than a technological enhancement to the payment system—they reshape the core mechanics of monetary policy transmission. Their successful integration will require adaptive policy design, continuous monitoring, and a forward-looking research agenda to safeguard both macroeconomic stability and financial system resilience in the digital currency era.

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# Appendix A: R Code for Econometric Analysis

Listing 1: R Code for Econometric Analysis

````
```{r}
# Packages Used
library(readxl)
library(dplyr)
library(lubridate)
library(fixest)       # fast, robust OLS with flexible vcov
library(sandwich)     # HAC vcov
library(lmtest)       # coeftest
library(modelsummary) # nice tables
library(broom)
library(janitor)
library(stringr)
library(ggplot2)
library(haven)
library(readr)
````
````

```
library(rmarkdown)
library(broom)
library(knitr)
library(dplyr)
library(Hmisc)
library(tidyr)
library(lmtest)
library(mfx)
library(rms)
library(plm)
library(AER)

# Code Chunk Formatting
opts_chunk$set(echo = TRUE)
options(digits = 3)
```

**Loading & Previewing Data**

```{r}
path <- "'/Users/anthonyrivera/Desktop/CBDC␣Merged␣Master␣Data␣Set.xlsx'"
data <- read_excel("/Users/anthonyrivera/Desktop/CBDC␣Merged␣Master␣Data␣Set.xlsx")
glimpse(data)
```

**Degree of Missingness Report:**

```{r}
data %>%
  summarise(across(everything(), ~mean(is.na(.)) * 100)) %>%
  pivot_longer(everything(), names_to = "Column", values_to = "MissingPercentage") %>%
  mutate(MissingPercentage = round(MissingPercentage, 2))
```

**Further Preperation Formatting **

```{r}
# Creating New Variable
data$Spread_Lend_Policy <- data$Change_LendingRate - data$MP_shock

# ---- HAC (Newey West) helper ----
hac6 <- function(m) sandwich::NeweyWest(m, lag = 6, prewhite = FALSE, adjust = TRUE)

# ---- Utility: 25bp pre/post effect from coefficients ----
pass_25bp <- function(model) {
```

```
  cf <- coef(model)
  theta <- unname(cf["MP_shock"])
  delta <- if ("MP_shock:CBDC_Pilot" %in% names(cf)) unname(cf["MP_shock:CBDC_Pilot"]) else
      NA_real_
  pre_25  <- 0.25 * theta
  post_25 <- 0.25 * (theta + delta)
  out <- data.frame(theta = theta, delta = delta,
                    pre_25bp = pre_25, post_25bp = post_25,
                    delta_effect_25bp = post_25 - pre_25)
  out
}

```

**Econometric Analysis**

  **Price Channel (rates & spread)**
  ```{r}
# ============================
# 1) PRICE CHANNEL (rates & spread)
# ============================

# Lending rate change
m_price_lend <- lm(
  Change_LendingRate ~ MP_shock + MP_shock:CBDC_Pilot +
    Core_InflationRates + CB_assets_GDP + GDP_Deflator_Index + UnemploymentRate + QE_Flow,
  data = data
)

# Deposit rate change (note the backticks for % in the name)
m_price_depo <- lm(
  `%Change_Average_Deposit_Rates` ~ MP_shock + MP_shock:CBDC_Pilot +
    Core_InflationRates + CB_assets_GDP + GDP_Deflator_Index + UnemploymentRate + QE_Flow,
  data = data
)

# Lending policy spread
m_price_spread <- lm(
  Spread_Lend_Policy ~ MP_shock + MP_shock:CBDC_Pilot +
    Core_InflationRates + CB_assets_GDP + GDP_Deflator_Index + UnemploymentRate + QE_Flow,
  data = data
```

```
)

cat("\n--- PRICE: Change_LendingRate (HAC) ---\n")
print(coeftest(m_price_lend,  vcov = hac6(m_price_lend)))
cat("\n25bp pre vs post:\n"); print(pass_25bp(m_price_lend))

cat("\n--- PRICE: %Change_Average_Deposit_Rates (HAC) ---\n")
print(coeftest(m_price_depo,  vcov = hac6(m_price_depo)))
cat("\n25bp pre vs post:\n"); print(pass_25bp(m_price_depo))

cat("\n--- PRICE: Spread_Lend_Policy (HAC) ---\n")
print(coeftest(m_price_spread, vcov = hac6(m_price_spread)))
cat("\n25bp pre vs post:\n"); print(pass_25bp(m_price_spread))

```

**Quantity / Credit Channel**

```{r}
# ============================
# 2) QUANTITY / CREDIT CHANNEL
# ============================

# Deposits level
m_qty_deposits <- lm(
  Deposits_Bank ~ MP_shock + MP_shock:CBDC_Pilot +
    Core_InflationRates + CB_assets_GDP + GDP_Deflator_Index + UnemploymentRate + QE_Flow,
  data = data
)

# Credit growth (% YoY)
m_qty_credit <- lm(
  CreditGrowth ~ MP_shock + MP_shock:CBDC_Pilot +
    Core_InflationRates + CB_assets_GDP + GDP_Deflator_Index + UnemploymentRate + QE_Flow,
  data = data
)

cat("\n--- QUANTITY: Deposits_Bank (HAC) ---\n")
print(coeftest(m_qty_deposits, vcov = hac6(m_qty_deposits)))
cat("\n25bp pre vs post:\n"); print(pass_25bp(m_qty_deposits))

cat("\n--- QUANTITY: CreditGrowth (HAC) ---\n")
```

```
print(coeftest(m_qty_credit,   vcov = hac6(m_qty_credit)))
cat("\n25bp pre vs post:\n"); print(pass_25bp(m_qty_credit))
```

**Exchange-Rate / Portfolio Channel **

```{r}
m_fx_neer <- lm(
  NEER ~ MP_shock + MP_shock:CBDC_Pilot +
    Core_InflationRates + CB_assets_GDP + GDP_Deflator_Index + UnemploymentRate + QE_Flow,
  data = data
)

cat("\n--- FX: NEER (HAC) ---\n")
print(coeftest(m_fx_neer, vcov = hac6(m_fx_neer)))
cat("\n25bp pre vs post:\n"); print(pass_25bp(m_fx_neer))
```

**Visualization**

```{r}
# Ensure CBDC_Pilot is a factor for coloring
data$CBDC_Pilot <- factor(data$CBDC_Pilot, labels = c("Pre-CBDC", "Post-CBDC"))

# ===== 1) PRICE CHANNEL: Change_LendingRate =====
ggplot(data, aes(x = MP_shock, y = Change_LendingRate, color = CBDC_Pilot)) +
  geom_point(alpha = 0.6) +
  geom_smooth(method = "lm", se = TRUE) +
  labs(
    title = "Price Channel: Lending Rate Pass-through",
    x = "Monetary Policy Shock (MP_shock)",
    y = "Change in Lending Rate",
    color = "Period"
  ) +
  theme_minimal()

# ===== 2) PRICE CHANNEL: %Change_Average_Deposit_Rates =====
ggplot(data, aes(x = MP_shock, y = `%Change_Average_Deposit_Rates`, color = CBDC_Pilot)) +
  geom_point(alpha = 0.6) +
  geom_smooth(method = "lm", se = TRUE) +
  labs(
    title = "Price Channel: Deposit Rate Pass-through",
    x = "Monetary Policy Shock (MP_shock)",
```

```
    y = "% Change in Average Deposit Rates",
    color = "Period"
  ) +
  theme_minimal()

# ===== 3) PRICE CHANNEL: Spread_Lend_Policy =====
ggplot(data, aes(x = MP_shock, y = Spread_Lend_Policy, color = CBDC_Pilot)) +
  geom_point(alpha = 0.6) +
  geom_smooth(method = "lm", se = TRUE) +
  labs(
    title = "Price Channel: L e n d i n g Policy Rate Spread",
    x = "Monetary Policy Shock (MP_shock)",
    y = "Spread: Lending Rate - Policy Rate",
    color = "Period"
  ) +
  theme_minimal()

# ===== 4) QUANTITY/CREDIT CHANNEL: Deposits_Bank =====
ggplot(data, aes(x = MP_shock, y = Deposits_Bank, color = CBDC_Pilot)) +
  geom_point(alpha = 0.6) +
  geom_smooth(method = "lm", se = TRUE) +
  labs(
    title = "Quantity Channel: Bank Deposits Response",
    x = "Monetary Policy Shock (MP_shock)",
    y = "Bank Deposits",
    color = "Period"
  ) +
  theme_minimal()

# ===== 5) QUANTITY/CREDIT CHANNEL: CreditGrowth =====
ggplot(data, aes(x = MP_shock, y = CreditGrowth, color = CBDC_Pilot)) +
  geom_point(alpha = 0.6) +
  geom_smooth(method = "lm", se = TRUE) +
  labs(
    title = "Credit Channel: Credit Growth Response",
    x = "Monetary Policy Shock (MP_shock)",
    y = "Credit Growth (%)",
    color = "Period"
  ) +
  theme_minimal()
```

```
# ===== 6) EXCHANGE RATE CHANNEL: NEER =====
ggplot(data, aes(x = MP_shock, y = NEER, color = CBDC_Pilot)) +
  geom_point(alpha = 0.6) +
  geom_smooth(method = "lm", se = TRUE) +
  labs(
    title = "Exchange Rate Channel: NEER Response",
    x = "Monetary Policy Shock (MP_shock)",
    y = "Nominal Effective Exchange Rate",
    color = "Period"
  ) +
  theme_minimal()
```